



# Chapter 10 - Variational Inference

## Quiz

Q1: For a given V.F.  $Q$  and a target distribution  $P$ , the optimal V.Approx.  $q$  is the distribution w/ smallest KL divergence, i.e.

$$q^* = \arg \min_{q \in Q} KL[q||p]$$

- we have a complex target distribution  $p(\theta)$  (e.g. posterior), from which we can't compute directly.

- we choose simpler family of dist.  $Q$

- we want the best approx  $q^*(\theta) \in Q$  to  $p(\theta)$

○ why it is TRUE?

- V.I turns inference into optimization

- it looks for the member of my V.F.  $Q$ , that gets as close as possible (KL div.) to the true posterior

○ minimizing KL • makes the approx.  $q(\theta)$  close to  $p(\theta)$  in prob. cond.

• Turns the inference into prob. of minimizing a scalar loss KL

○ KL div. from  $q(\theta)$  to  $p(\theta)$  is:

$$KL[q(\theta)||p(\theta)] = \int q(\theta) \log \frac{q(\theta)}{p(\theta)} d\theta$$

- it measures how much extra info. you need if you use  $q$  instead of the true dist.  $P$ .

Q2: The KL-div for the best approx.  $KL[q^*||p]$ , where  $q^* = \arg \min_{q \in Q} KL[q||p]$  is 0 if the target dist.  $p$  belongs to the var. fam. TRUE

○ Why? • KL div. between 2 dist. is 0 iff the 2 dist. are equal almost everywhere.

$$KL[q||p] = 0 \Leftrightarrow q = p$$

• If  $p \in Q$ , then  $\exists q \in Q$  s.t.  $q = p$

• the optima.  $q^* = \arg \min_{q \in Q} KL[q||p]$  will choose  $q = p$ , bcs. that results in KL div = 0

Q3: minimum  $KL[q||p]$  w.r.t.  $q$  leads to the same sol. as min.  $KL[p||q]$  w.r.t.  $q$ .  $\Rightarrow$  False !

○ KL is asymmetric  $KL[q||p] \neq KL[p||q] \Rightarrow$  minimizing each gives different result.

Q4: if we change the approx.  $q$  s.t. the KL div  $KL[q||p]$  increases, the approx.  $q$  becomes a better approx of  $p \Rightarrow$  FALSE !

○  $KL[q||p]$  is a measure of how dif. the dist.  $q$  is from the dist.  $p$

• smaller KL  $\Rightarrow q$  is closer to  $p$

• larger KL  $\Rightarrow q$  is away from  $p$

• as KL increases, the approximation  $q$  is getting worse, not better

Q5: The contribution to the KL div. is generally large in regions, where the  $q$  is small and  $p$  is large

$$KL[q||p] = \int q(z) \ln \left[ \frac{q(z)}{p(z)} \right] dz \Rightarrow \text{FALSE}$$

- we are integrating over values  $z$  and weighing each pt. by  $q(z)$   $\Rightarrow$  not by  $p(z)$

- if  $q$  is very small, then that region contributes almost nothing to the KL divergence  $\Rightarrow$  regardless of  $p(z)$

- if  $q(z) = 0.001$   $\wedge$   $p(z) = 0.8$  because we multiply small # by a log

- Q6: The contribution to the KL div. is generally large in regions, where the  $q$  is large and  $p$  is small

$$KL[q||p] = \int q(z) \ln \left[ \frac{q(z)}{p(z)} \right] dz \Rightarrow \text{TRUE}$$

- if  $q(z)$  is large, the region gets a lot of weight
  - if  $q(z) \gg p(z)$ , then  $\log \left( \frac{q}{p} \right)$  is positive and large
- $\Rightarrow$  their product is large and positive

- Q7: enlarging the V.F.  $Q$  leads to more accurate prediction.  $\Rightarrow$  True

- enlarging  $Q$  means we allow for flexible distributions. Going from mean-field Gaussian <sup>(indep. marginals)</sup>  $\Rightarrow$  to a full covariance Gaussian. Or to mixtures of Gaussians

$$q^* = \arg \min_{q \in Q} KL[q||p]$$

- increasing  $Q$ , we expand the search space  $\Rightarrow$  so we are most likely to find a  $q^*$ , that gets closer to  $p$ .

- Q8: Mean-field V.F. works better because they capture correlation in posterior.  $\Rightarrow$  FALSE

M-F assumes all variables are  $\perp$ .

$$q(z) = \prod_i q_i(z_i)$$

$\Rightarrow$  faster optimization, but does not capture correlations between variables in the posterior.

- if the true posterior has strong correlations, the mean-field approx is POOR.
- $\Rightarrow$  more expressive families (full cov Gauss) are better

- Q9: M-F V-F often lead to faster algo, bcs. it ignores the posterior correlation.  $\Rightarrow$  TRUE

- each latent variable can be updated independently.

$\Rightarrow$  simpler update rules

$\Rightarrow$  no need to compute or optimize a full cov. mat

$\Rightarrow$  often even closed-form updates

# Minimizing the KL divergence

$$KL[q||P(D)] = \int q(z) \ln \frac{q(z)}{p(z|D)} dz = \mathbb{E}_q[\ln q(z)] - \mathbb{E}_q[\ln p(z|D)]$$

• how we replace following term  $\mathbb{E}_q[\ln p(z|D)]$  w/  $\mathbb{E}_q[\ln p(z, D)]$ , we use  $\frac{p(z|D) \cdot p(D)}{p(D)}$

$$= \mathbb{E}_q[\ln p(z)] - \mathbb{E}_q[\ln p(z|D)]$$

$$= \mathbb{E}_q[\ln p(z)] - \mathbb{E}_q\left[\ln \frac{p(D, z)}{p(D)}\right]$$

$$= \mathbb{E}_q[\ln p(z)] - \left[ \mathbb{E}_q[\ln p(D, z) - p(D)] \right]$$

$$= \mathbb{E}_q[\ln p(z)] - \mathbb{E}_q[\ln p(D, z)] + \mathbb{E}_q[\ln p(D)] \quad \text{notice the sign flip}$$

$$KL[q||p] = \mathbb{E}_q[\ln p(z)] - \mathbb{E}_q[\ln p(D, z)] + \ln p(D)$$

○ Rearrange

$$\ln p(D) = \underbrace{\mathbb{E}_q[\ln p(D, z)] - \mathbb{E}_q[\ln p(z)]}_{\text{ELBO} = L[q]} + KL[q||p]$$

•  $KL[q||p] \geq 0$  and ELBO is a lower bound on  $\ln p(D)$

$$L[q] \equiv \mathbb{E}_q[\ln p(D, z)] - \mathbb{E}_q[\ln q(z)] \leq \ln p(D)$$

By def:  $\mathbb{E}_q[f(z)] = \int q(z) \cdot f(z) dz$

$$\therefore L[q] = \int q(z) \cdot \ln p(D, z) dz - \int q(z) \cdot \ln q(z) dz$$

→ now we assume the mean-factorization  $q(z) = \prod_{i=1}^J q(z_i)$

$$L[q] = \int \left( \prod_{i=1}^J q(z_i) \right) \ln p(D, z) dz - \int \left( \prod_{i=1}^J q(z_i) \right) \ln \left( \prod_{i=1}^J q(z_i) \right) dz$$



1.2 Show that  $\ln q^*(u) = u(N\bar{x} + \lambda_0 \mu_0) \mathbb{E}_{q(\gamma)}[\tilde{\gamma}] - \frac{1}{2} u^2 (N + \lambda_0) \mathbb{E}_{q(\gamma)}[\tilde{\gamma}] + C$

$$\ln q^*(u) =$$